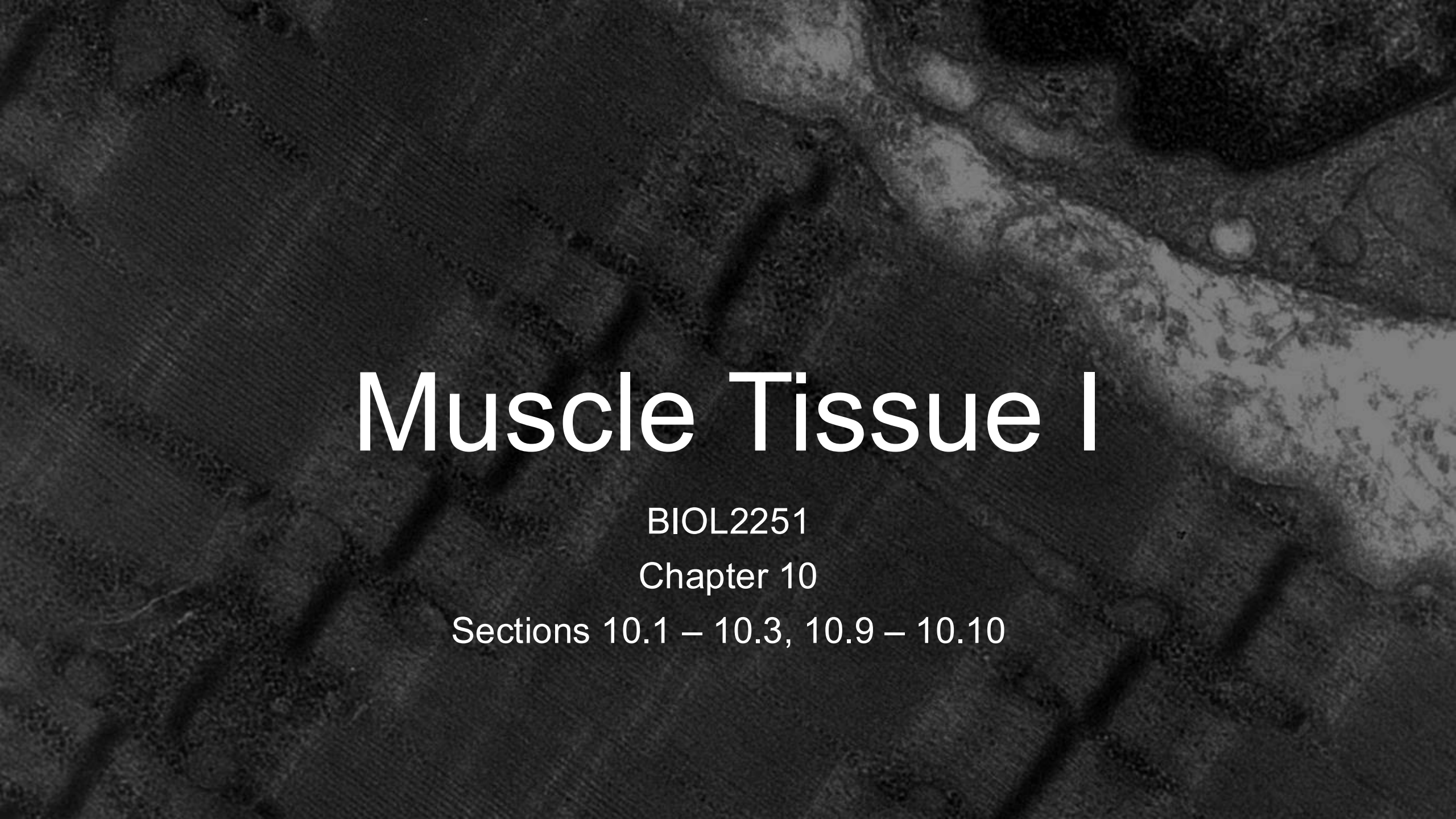


A grayscale electron micrograph of muscle tissue, showing the characteristic striated pattern of myofibrils with alternating light and dark bands. The image is used as a background for the title slide.

Muscle Tissue I

BIOL2251

Chapter 10

Sections 10.1 – 10.3, 10.9 – 10.10

Lecture Quiz



Test what you know!

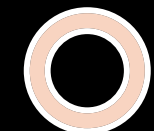


Quiz auto submits at **12:55 PM**



Password = **myofibril**

No notes – just what is stored in your brain



Properties and Functions Brain Dump

Properties and Functions Brain Dump

- **Instructions**

- On your own, **write down the key functions and properties of muscle tissue**
 - Compare your list with a partner
 - I will call on students around the room to contribute

- **Bloom's taxonomy = Remember**

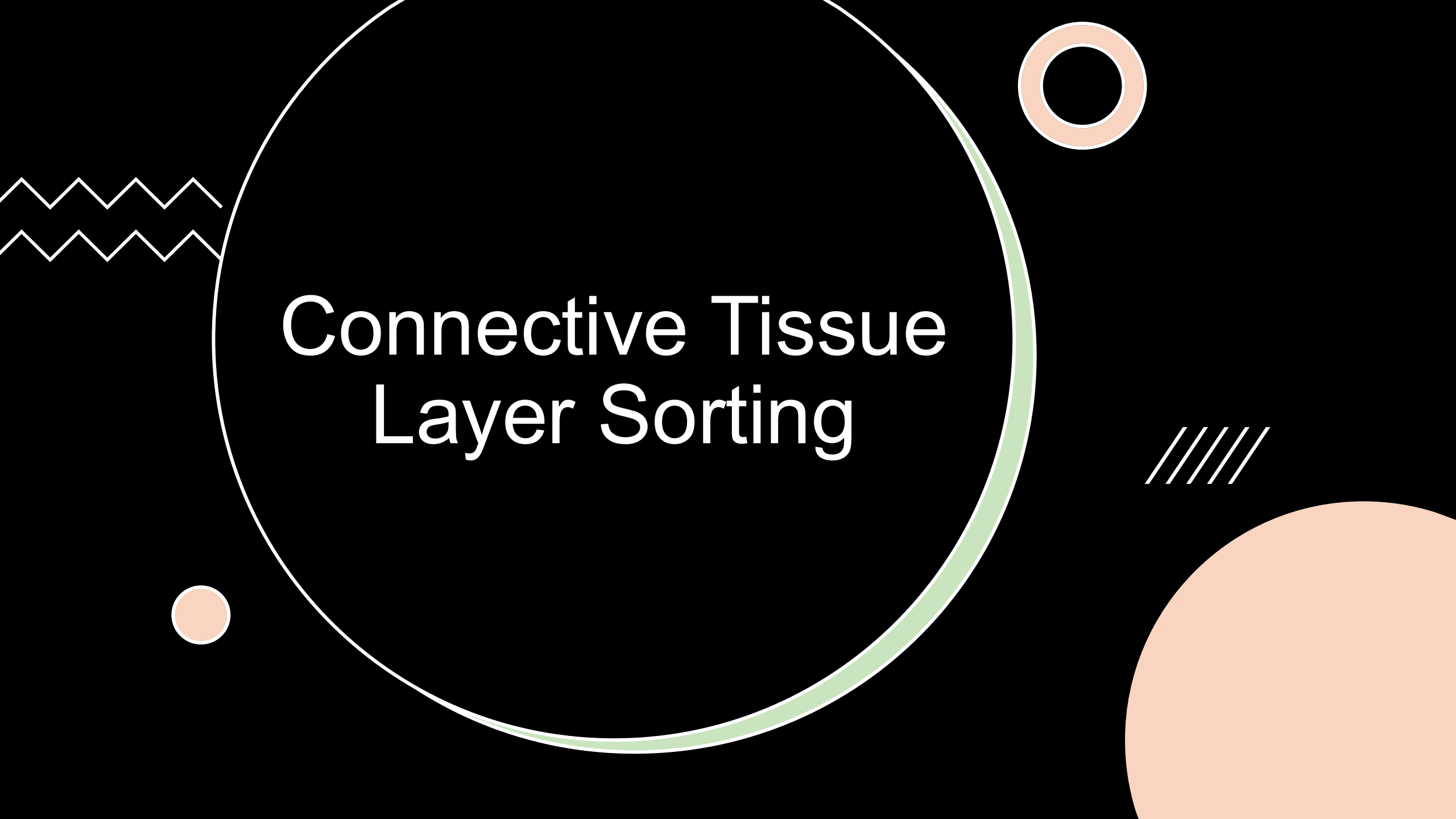
Properties and Functions Brain Dump

Functions

- Producing movement
- Maintaining posture and body position
- Supporting soft tissues
- Guarding body entrances and exits
- Maintaining body temperature
- Storing nutrients

Properties

- Excitability
- Contractility
- Extensibility
- Elasticity



Connective Tissue Layer Sorting

Connective Tissue Layer Sorting

- **Instructions**

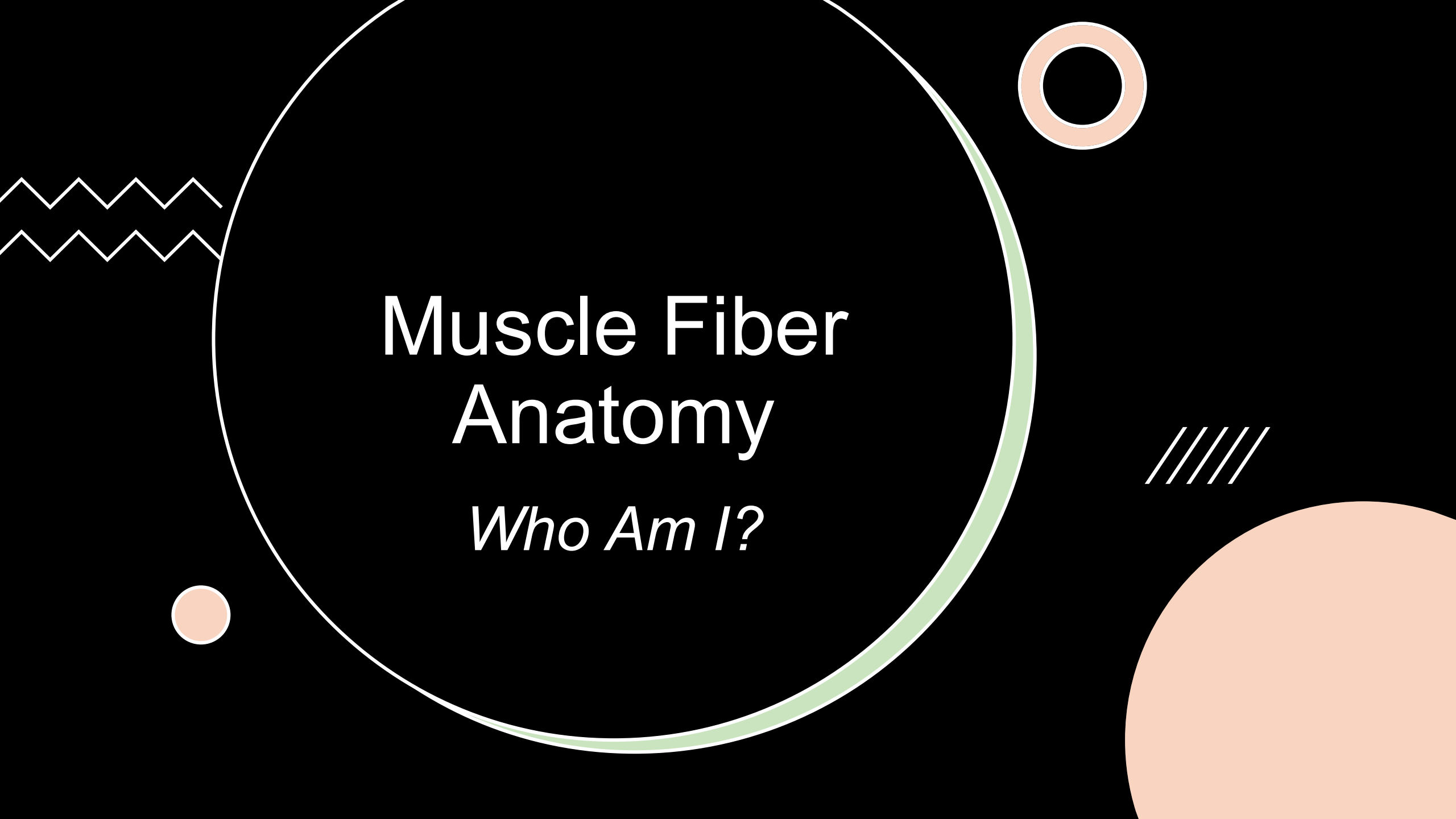
- Individually match the connective tissue layer names to their descriptions
 - Match names to location and function
 - Compare with a neighbor once you have your matches
 - I will use a random number to pick a table to share their work
- **Bloom's taxonomy = Remember, Understand**

Connective Tissue Layer Sorting

Layer Name	Location	Function
Epimysium	Surrounds individual muscle fibers	Provides passage for nerves and blood vessels to reach fascicles; groups fibers into functional bundles
Perimysium	Surrounds the entire skeletal muscle	Provides a pathway for capillaries and nerve fibers to reach each individual muscle fiber
Endomysium	Surrounds bundles of muscle fibers (fascicles)	Protects the muscle; anchors muscle to tendons and adjacent structures

Connective Tissue Layer Sorting

Layer Name	Location	Function
Epimysium	Surrounds the entire skeletal muscle	Protects the muscle; anchors muscle to tendons and adjacent structures
Perimysium	Surrounds bundles of muscle fibers (fascicles)	Provides passage for nerves and blood vessels to reach fascicles; groups fibers into functional bundles
Endomysium	Surrounds individual muscle fibers	Provides a pathway for capillaries and nerve fibers to reach each individual muscle fiber



Muscle Fiber Anatomy

Who Am I?

Muscle Fiber Anatomy – Who Am I?

- **Instructions**

- Each clue describes a specific structure found in a skeletal muscle fiber
 - Write the name of the muscle fiber on your notecard
 - Volunteers who provide the correct structure get a prize
- **Bloom's taxonomy = Remember, Understand**

Muscle Fiber Anatomy – Who Am I?

1. I am the plasma membrane of a muscle fiber. I wrap the entire cell and play a key role in transmitting electrical signals.
2. I am a network of interconnected protein tubes that dip deep into the interior of the muscle fiber. I allow an electrical signal generated at the surface to reach the very center of the cell almost simultaneously.
3. I am the muscle fiber's version of smooth ER. I form a network of membranous sacs around each myofibril, and I am the most important storage site for a critical ion needed to trigger contraction.

Muscle Fiber Anatomy – Who Am I?

4. I am long, thread-like structures that run the entire length of the muscle fiber. I am made of repeating contractile units, and I am what actually shortens during contraction.

5. I am the thicker of the two main types of myofilaments. I am composed of a protein with a head and a tail, and my heads are the 'motors' of muscle contraction.

6. I am the thinner of the two main types of myofilaments. I am primarily made of a globular protein arranged in a double helix, and I serve as the track that the thick filament heads walk along.

Muscle Fiber Anatomy – Who Am I?

Clue	Structure	Key Detail
1	Sarcolemma	Plasma membrane of muscle fiber; transmits action potentials
2	Transverse (T) tubules	Extensions of sarcolemma that penetrate deep into fiber; ensure synchronous contraction
3	Sarcoplasmic reticulum (SR)	Stores and releases Ca^{2+} ; Ca^{2+} release triggers contraction
4	Myofibrils	Cylindrical contractile organelles; composed of sarcomeres in series
5	Thick filaments (myosin)	Myosin heads form cross-bridges with actin; ATPase activity powers contraction
6	Thin filaments (actin)	F-actin double helix; binding sites for myosin heads



Sarcomere Band Analysis

Sarcomere Band Analysis: Part 1

- **Instructions**

- Work on your own (at first)
- You will be given a description of sarcomere anatomy
 - **Match each description to the correct sarcomere structure**
- Check your work at your table before whole-class check
- **Bloom's taxonomy = Understand, Apply**

Sarcomere Band Analysis: Part 1

Match each description to the correct sarcomere structure (Z line, M line, A band, I band, H zone):

- The boundary of a sarcomere; thin filaments are anchored here.
- The central line of the sarcomere; anchors thick filaments in the center.
- The zone that contains *only* thick filaments (no overlap with thin filaments in a relaxed fiber).
- The zone that spans the entire length of thick filaments, regardless of contraction state.
- The zone containing *only* thin filaments — no thick filaments present.

Sarcomere Band Analysis: Part 1

- The boundary of a sarcomere; thin filaments are anchored here = **Z line**
- The central line of the sarcomere; anchors thick filaments in the center = **M line**
- The zone that contains *only* thick filaments (no overlap with thin filaments in a relaxed fiber) = **H zone**
- The zone that spans the entire length of thick filaments, regardless of contraction state = **A band**
- The zone containing *only* thin filaments — no thick filaments present = **I band**

Sarcomere Band Analysis: Part 2

- **Instructions**

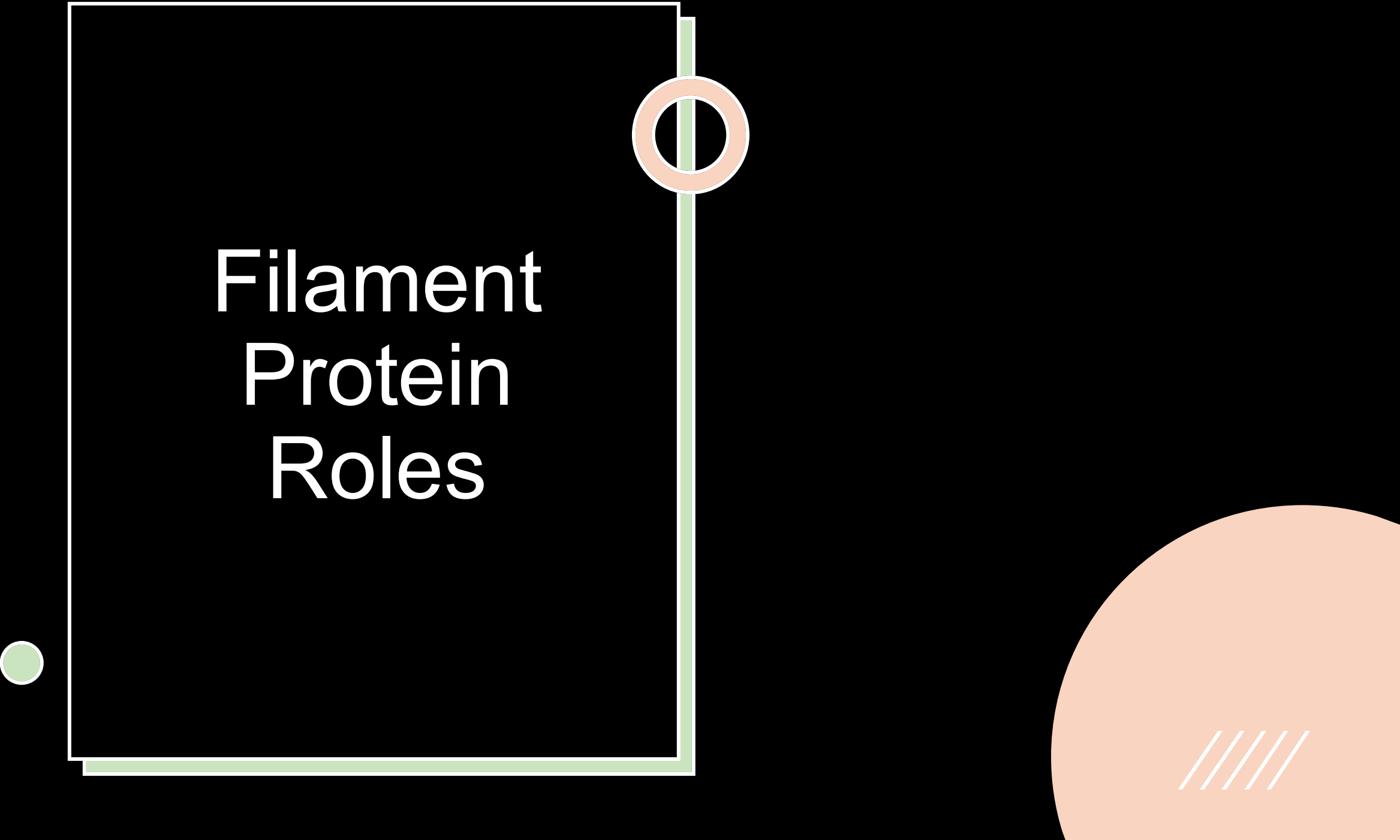
- Work in small groups
- **Predict how each structure changes during contraction**
 - **Indicate whether the structure gets shorter, stays the same, or gets longer**
 - Provide a one sentence justification to explain why
 - I will use a random number to pick a table to share their work
- **Bloom's taxonomy = Understand, Apply**

Sarcomere Band Analysis: Part 2

Structure	Change?	Why?
I band		
A band		
H zone		
Sarcomere (overall)		
Thick filament length		
Thin filament length		

Sarcomere Band Analysis: Part 2

Structure	Change?	Why?
I band	Gets shorter	Thin filaments slide inward toward M line, reducing the thin-filament-only zone
A band	Stays the same	Width equals thick filament length, which does not change
H zone	Gets shorter	Thin filaments overlap more with thick filaments, reducing the thick-only zone
Sarcomere (overall)	Gets shorter	Z lines are pulled closer together as thin filaments slide inward
Thick filament length	Stays the same	Filaments themselves don't shorten — sliding occurs, not compression
Thin filament length	Stays the same	Same as above — sliding, not compression



Filament Protein Roles

Filament Protein Roles

- **Instructions**

- Work in small groups
- You will work through a set of three mini-scenarios describing a disruption to a specific filament protein
 - **Identify which protein is affected and explain what goes wrong structurally or functionally – and why**
- I will use a random number to pick a table to share their work

- **Bloom's taxonomy = Understand, Apply**

Filament Protein Roles

- **Scenario A:** A researcher develops a drug that destroys a specific protein in skeletal muscle fibers. Even when the muscle is completely at rest and Ca^{2+} levels are very low, the myosin-binding sites on actin remain permanently exposed. What protein was lost, what was its normal job, and what structural consequence does its absence create?

Filament Protein Roles

- Lost protein: **Tropomyosin**
- Normal role: Tropomyosin is a protein that runs along the groove of the actin double helix and physically covers the myosin-binding sites on actin at rest, preventing unwanted interaction between actin and myosin
- Consequence of absence: With no tropomyosin to block binding sites, actin is permanently accessible regardless of Ca^{2+} levels — the regulatory off switch is gone entirely

Filament Protein Roles

- **Scenario B:** A genetic mutation prevents troponin from changing shape when Ca^{2+} binds to it. The protein can still bind Ca^{2+} normally, but its shape does not change afterward. What is the normal consequence of Ca^{2+} binding to troponin, and what goes wrong in this patient?

Filament Protein Roles

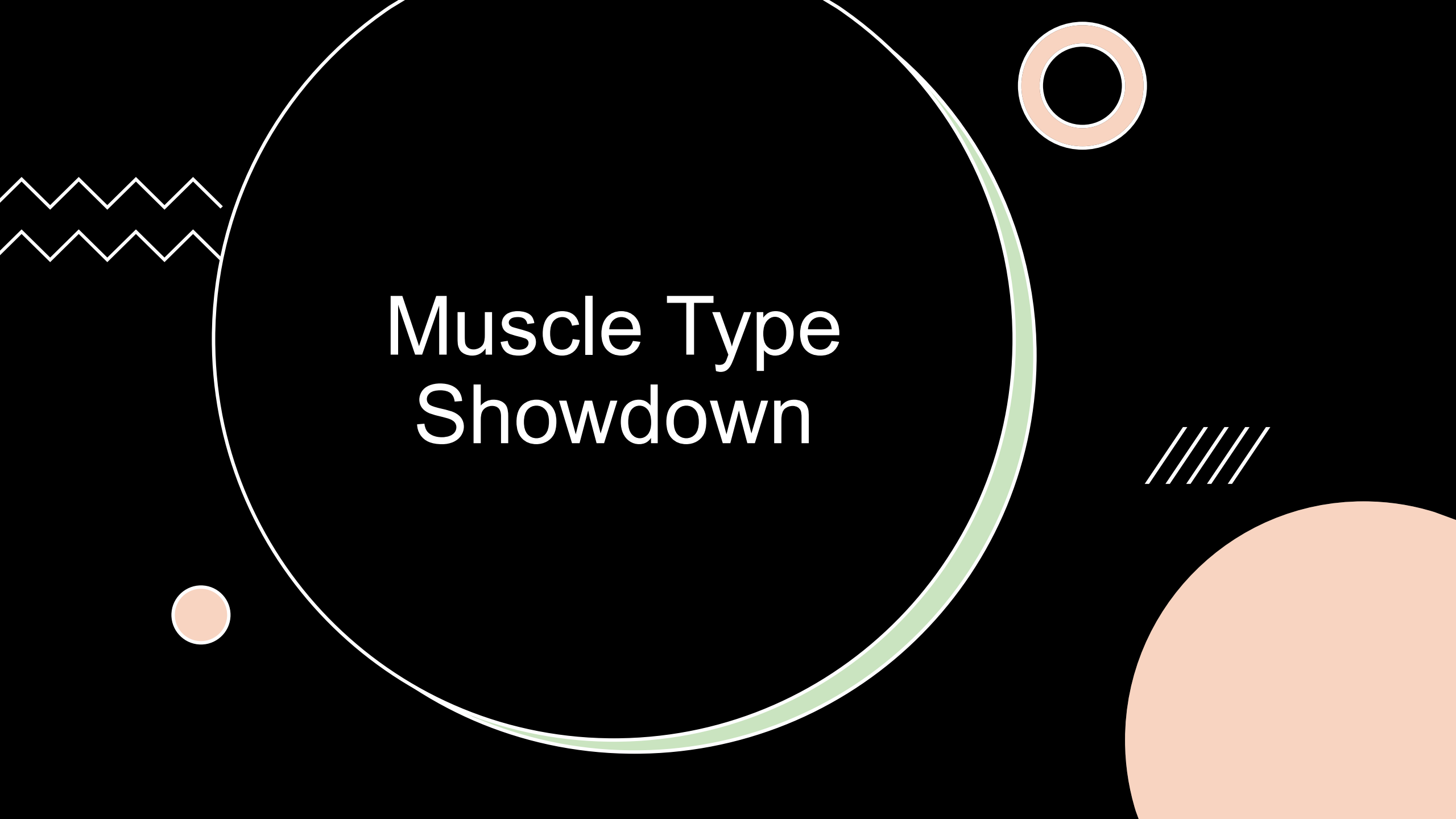
- Normal sequence: Ca^{2+} binds to **troponin** → troponin changes shape (conformational change) → pulls tropomyosin away from the myosin-binding sites on actin → binding sites become exposed
- In this patient: Ca^{2+} can bind troponin, but since troponin's shape doesn't change, tropomyosin is never moved → binding sites on actin remain covered → the thin filament is permanently in its off state even when Ca^{2+} is present

Filament Protein Roles

- **Scenario C:** A patient has a rare disease in which titin is completely absent from their sarcomeres. A physical therapist notices that the patient's muscles are damaged very easily when passively stretched. Using titin's structural role, explain why this makes sense.

Filament Protein Roles

- **Titin** is an elastic protein that runs from the Z line to the M line, anchoring thick filaments in the center of the sarcomere and acting as a molecular spring
- Without titin, thick filaments are no longer tethered to the Z line → when the sarcomere is passively stretched, there is no elastic element to resist or absorb the force → sarcomere structure becomes unstable and vulnerable to mechanical damage during stretching
- Titin also ensures thick filaments remain centered in the sarcomere; its absence would allow filaments to shift out of alignment

The background is black with several abstract elements: a large circle with a thin white outline and a thick light green inner ring; two horizontal white zigzag lines on the left; a small light orange circle in the bottom left; a larger light orange circle in the bottom right; a light orange ring in the top right; and five parallel white diagonal lines on the right side.

Muscle Type Showdown

Muscle Type Showdown

- **Instructions**

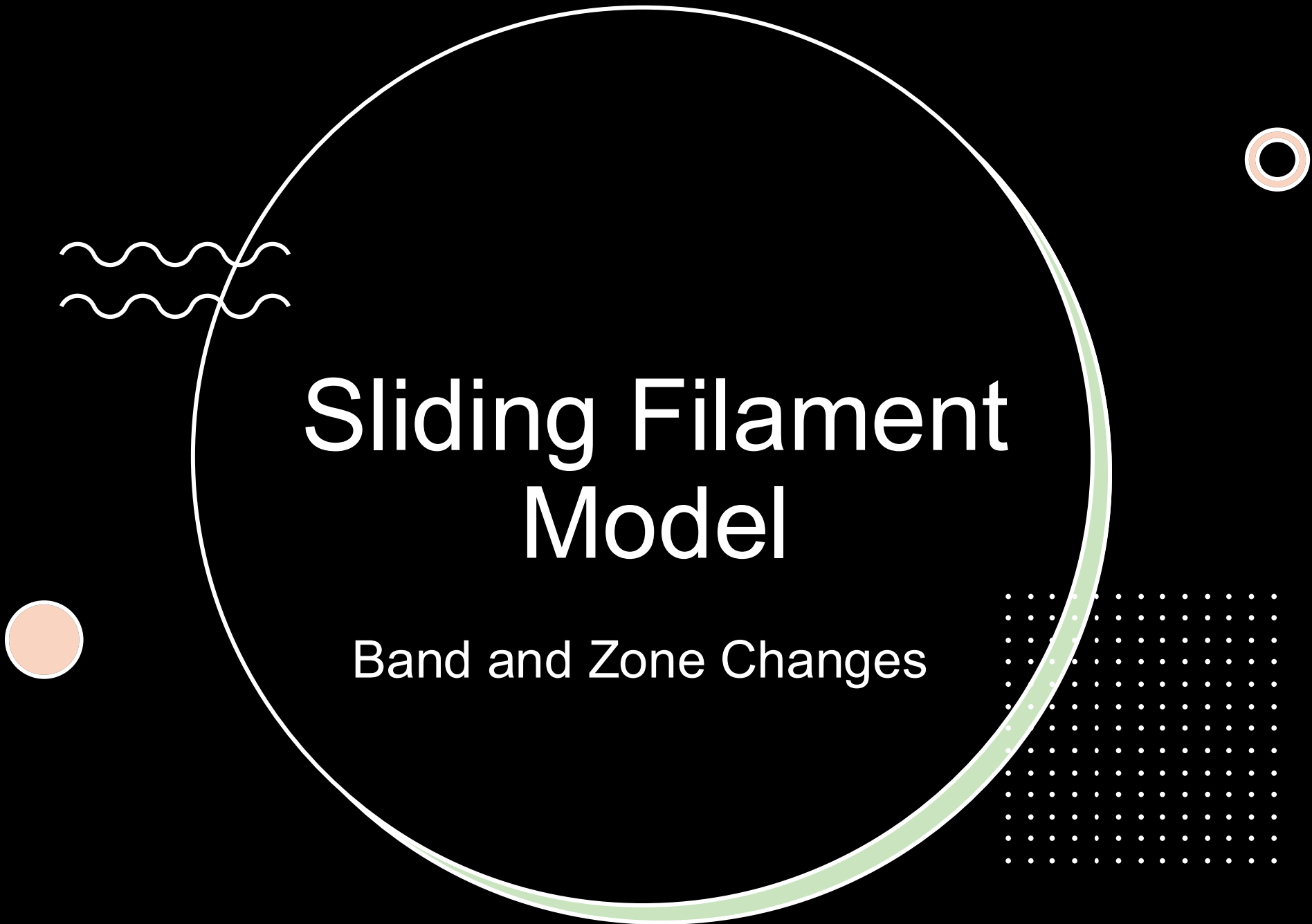
- Work in small groups
- **Complete the comparison table for the three muscle types**
 - We will complete the table as a class
- All groups should contribute
- **Bloom's taxonomy = Analyze**

Muscle Type Showdown

Feature	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Cell shape			
Nuclei (number/location)			
Striated? (Yes/No)			
Voluntary or involuntary control			
Presence of T-tubules			
Intercalated discs?			
Speed and fatigue resistance			
Location in body			

Muscle Type Showdown

Feature	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Cell shape	Long, cylindrical fibers	Branched, short cells	Spindle-shaped (fusiform)
Nuclei (number/location)	Multiple, peripherally located	Single (or occasionally 2), centrally located	Single, centrally located
Striated? (Yes/No)	Yes	Yes	No
Voluntary or involuntary control	Voluntary	Involuntary (autorhythmic)	Involuntary
Presence of T-tubules	Yes (well-developed)	Yes (present, wider)	Absent (or rudimentary)
Intercalated discs?	No	Yes — contain gap junctions and desmosomes	No
Speed and fatigue resistance	Fast; fatigues	Moderate; highly fatigue resistant	Slow; very fatigue resistant
Location in body	Attached to skeleton	Heart walls	Walls of hollow organs, blood vessels, airways



Sliding Filament Model

Band and Zone Changes

Sliding Filament Model

- **Instructions**

- Work in pairs
- I will present a set of statements describing what happens to a sarcomere during muscle contraction
- **For each statement, mark it as TRUE or FALSE**
 - **If a statement is false, rewrite it to make it correct**
- **Bloom's taxonomy = Apply, Evaluate**

Sliding Filament Model

- During contraction, the I band gets shorter because thin filaments slide toward the M line.
- The A band shortens during contraction because thick filaments are pulled inward.
- The H zone narrows during contraction because thin filaments increasingly overlap with thick filaments.
- The Z lines move closer together during contraction, which is why the overall sarcomere shortens.
- Thick filaments and thin filaments both physically shorten during contraction.
- Even at maximum contraction, the A band width remains unchanged.
- The H zone would completely disappear if thin filaments overlapped entirely across the center of the sarcomere.

Sliding Filament Model

- During contraction, the I band gets shorter because thin filaments slide toward the M line = **TRUE**
- The A band shortens during contraction because thick filaments are pulled inward = **FALSE**
- The H zone narrows during contraction because thin filaments increasingly overlap with thick filaments = **TRUE**
- The Z lines move closer together during contraction, which is why the overall sarcomere shortens = **TRUE**
- Thick filaments and thin filaments both physically shorten during contraction = **FALSE**
- Even at maximum contraction, the A band width remains unchanged = **TRUE**
- The H zone would completely disappear if thin filaments overlapped entirely across the center of the sarcomere = **TRUE**

Sliding Filament Model

- Continue working in your pairs to respond to these justification questions:
 1. A student says: The sarcomere shortens, so the filaments must be getting compressed or folded. Using what you know about sliding filament mechanics, explain what is actually happening and why the student's reasoning is incorrect.
 2. The I band and H zone both shorten during contraction, but for different reasons. Explain what each zone represents structurally and use that to explain why each one narrows.

Sliding Filament Model

Justification 1

- Sarcomere shortens not because filaments compress, but because thin filaments *slide* past thick filaments toward the M line
- The filaments themselves stay the same length throughout contraction – only their degree of overlap changes

Sliding Filament Model

- Justification 2
- The **I band** contains only thin filaments – no thick filaments
 - It narrows because the thin filaments slide inward, reducing the region where thin filaments exist without any thick filament overlap
- The **H zone** contains only thick filaments – no thin filaments
 - It narrows because thin filaments slide into that previously unoccupied central region, reducing the thick-filament-only zone
- Both zones shrink, but one is being encroached upon by thin filaments moving *away from the Z line* (I band), while the other is being encroached upon by thin filaments moving *toward the M line* (H zone)